

Department of Electronic & Telecommunication Engineering, University of Moratuwa, Sri Lanka.

Assignment 2

220163X Fernando D.S.

Submitted in partial fulfillment of the requirements for the module

EN3330 Introduction to Engineering Optimization

2nd September 2026

1. $f(x) = \frac{1}{4} x_1^4 + \frac{b}{3} x_2^3 + b \ln(1 + x_3^2)$

(a) $$\nabla f(x) = \begin{bmatrix} x_1^3 \\ b x_2^2 \\ \frac{b}{(1 + x_3^2)} \cdot 2 x_3 \end{bmatrix}$$

when $b = 0.001$

$$\nabla f(x) = \begin{bmatrix} x_1^3 \\ 0.001 x_2^2 \\ \frac{0.002 x_3}{(1 + x_3^2)} \end{bmatrix}$$

x_1 dominates the contribution to the gradient when $b = 0.001$.

Since x_1 dominates the contribution to the gradient, x_1 changes rapidly while the changes in x_2 & x_3 are small.

(b.) $x^{(0)} = \begin{pmatrix} 2 \\ 2 \\ 2 \end{pmatrix}$ $\eta = 0.1$ $b = 0.001$

$$\nabla f(x) = \begin{bmatrix} x_1^3 \\ 0.001 x_2^2 \\ \frac{0.002 x_3}{(1 + x_3^2)} \end{bmatrix}$$

$$x^{(1)} = \begin{bmatrix} 2 \\ 2 \\ 2 \end{bmatrix} - 0.1 \begin{bmatrix} 8 \\ 0.004 \\ \frac{0.004}{5} \end{bmatrix}$$

$$= \begin{bmatrix} 1.2 \\ 1.9996 \\ 1.99992 \end{bmatrix}$$

$$x^{(2)} = \begin{bmatrix} 1.2 \\ 1.9996 \\ 1.99992 \end{bmatrix} - 0.1 \begin{bmatrix} 1.2^3 \\ 0.001 \times 1.9996^2 \\ \frac{0.002 \times 1.99992}{1 + 1.99992^2} \end{bmatrix}$$

$$= \begin{bmatrix} 1.0272 \\ 1.9992 \\ 1.9998 \end{bmatrix}$$

$$x^{(3)} = \begin{bmatrix} 1.0272 \\ 1.9992 \\ 1.9998 \end{bmatrix} - 0.1 \begin{bmatrix} 1.0272^2 \\ 0.001 \times 1.9992^2 \\ \frac{0.002 \times 1.9998}{1 + 1.9998^2} \end{bmatrix}$$

$$= \begin{bmatrix} 0.92168 \\ 1.9988 \\ 1.9997 \end{bmatrix}$$

(c.) Rate of change of x_1 is high where as the rate of change x_2 & x_3 is extremely small.

Since the scaling between the variables are a lot different using the same

learning rate is not appropriate here.

(d.) No. Standard gradient descent with a fixed learning rate is not suitable here.

Having 3 different learning rates chosen according to the contribution to the gradient is appropriate here.

If $\eta_1 \rightarrow$ learning rate of x_1

$\eta_2 \rightarrow$ " " " x_2

$\eta_3 \rightarrow$ " " " x_3

Based on the scaling

$$\eta_1 < \eta_2$$

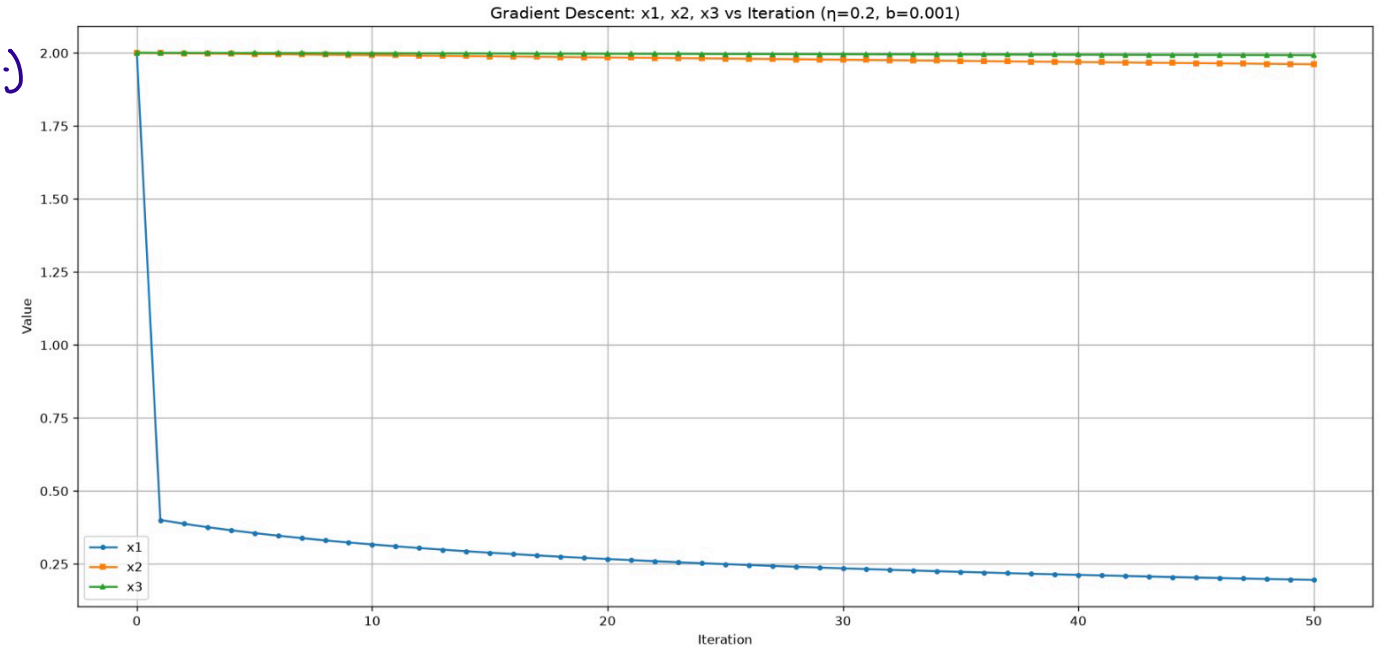
$$\eta_1 < \eta_3$$

should be like this

$$x_i^{(k+1)} = x_i^{(k)} - \eta_i \nabla f_i^{(k)}.$$

Because otherwise step size of x_1 is a lot higher compared to x_2 & x_3.

(e.)



Here in the 1st iteration the value of x_1 drops significantly. Afterwards since the value of x_1 is small it slowly converges to zero.

An adaptive learning can be used here where η_1 is small compared η_2 & η_2.

2./ $f(x) = (1-x_1)^2 + b(x_2 - x_1^2)^2$

(a.) $\nabla f(x) = \begin{bmatrix} -2(1-x_1) - 4b(x_2 - x_1^2)x_1 \\ 2b(x_2 - x_1^2) \end{bmatrix}$

$\nabla f(x) = \begin{bmatrix} -2 + 2x_1 - 4b x_1 x_2 + 4b x_1^3 \\ 2b(x_2 - x_1^2) \end{bmatrix}$

$\nabla^2 f(x) = \begin{bmatrix} 2 - 4b(x_2 - x_1^2) - 4b x_1(-2x_1) & -4b x_1 \\ 2b(-2x_1) & 2b \end{bmatrix}$

$= \begin{bmatrix} 2 - 4bx_2 + 4bx_1^2 + 8bx_1^2 & -4bx_1 \\ -4bx_1 & 2b \end{bmatrix}$

$$= \begin{bmatrix} 2 - 4bx_2 + 12bx_1^2 & -4bx_1 \\ -4bx_1 & 2b \end{bmatrix}$$

(b.) when $b=1$,

$$\nabla f(x) = \begin{bmatrix} -2 + 2x_1 - 4x_1x_2 + 4x_1^3 \\ 2x_2 - 2x_1^2 \end{bmatrix}$$

$$\nabla^2 f(x) = \begin{bmatrix} 2 - 4x_2 + 12x_1^2 & -4x_1 \\ -4x_1 & 2 \end{bmatrix}$$

$$x^{k+1} = x^k - [\nabla^2 f(x)]^{-1} \cdot \nabla f(x)$$

$$x^{(1)} = \begin{bmatrix} -1.2 \\ 1 \end{bmatrix} - \begin{bmatrix} 2 - 4 + 12 \times (-1.2)^2 & 4 \times 1.2 \\ 4 \times 1.2 & 2 \end{bmatrix}^{-1} \begin{pmatrix} -6.512 \\ -0.88 \end{pmatrix}$$

$$= \begin{bmatrix} -1.2 \\ 1 \end{bmatrix} - \begin{pmatrix} 15.28 & 4.8 \\ 4.8 & 2 \end{pmatrix}^{-1} \begin{pmatrix} -6.512 \\ -0.88 \end{pmatrix}$$

$$x^{(1)} = \begin{bmatrix} -0.029 \\ -1.368 \end{bmatrix} //$$

$$\nabla^2 f(x)^{(1)} = \begin{pmatrix} 2 + 4 \times 1.368 + 12 \times 0.029^2 & 4 \times 0.029 \\ 4 \times 0.029 & 2 \end{pmatrix}$$

$$\nabla f(x)^{(1)} = \begin{bmatrix} -2 - 2\times 0.029 - 4(-0.029)(-1.368) + 4(-0.029)^3 \\ 2(-1.368) - 2(-0.029)^2 \end{bmatrix}$$

$$x^{(2)} = \begin{bmatrix} -0.029 \\ -1.368 \end{bmatrix} - \begin{bmatrix} 7.482 & 0.116 \\ 0.116 & 2 \end{bmatrix}^{-1} \begin{bmatrix} -2.216 \\ -2.737 \end{bmatrix}$$

$$= \begin{bmatrix} 0.246 \\ -0.015 \end{bmatrix} //$$

$x_1 \rightarrow -1.2 \rightarrow -0.029 \rightarrow 0.246$

$x_2 \rightarrow 1 \rightarrow -1.368 \rightarrow -0.015$

first approximation moves farther away from the minimum but then a good improvement in the 2nd iteration. But the approximation is slowly getting towards the minimum.

(c.) when $b = 1000$,

$$\nabla f(x) = \begin{bmatrix} -2 + 2x_1 - 4b x_1 x_2 + 4b x_1^3 \\ 2b(x_2 - x_1^2) \end{bmatrix}$$

$$= \begin{bmatrix} -2 + 2x_1 - 4000 x_1 x_2 + 4000 x_1^3 \\ 2000(x_2 - x_1^2) \end{bmatrix}$$

$$= \begin{bmatrix} -2116.4 \\ -880 \end{bmatrix}$$

$$\nabla^2 f(x) = \begin{bmatrix} 2 - 400x_2 + 1200x_1^2 & -400x_1 \\ -400x_1 & 200 \end{bmatrix}$$

$$= \begin{bmatrix} 2 - 4000 + 12000(-1.2)^2 & -4000(-1.2) \\ -4000(-1.2) & 2000 \end{bmatrix}$$

$$= \begin{bmatrix} 13282 & 4800 \\ 4800 & 2000 \end{bmatrix}$$

$$x^{(1)} = \begin{bmatrix} -1.2 \\ 1 \end{bmatrix} - \begin{bmatrix} 13282 & 4800 \\ 4800 & 2000 \end{bmatrix}^{-1} \begin{bmatrix} -2116.4 \\ -880 \end{bmatrix}$$

$$= \begin{bmatrix} -1.1975 \\ 1.434 \end{bmatrix}$$

Here the hessian is changing rapidly. Therefore this approach is not effective. since a small error can affect significantly. Can avoid this

by quasi newton methods.

(d.) $f(x) = \frac{1}{2} x^T S x$

$\nabla f(x) = \frac{1}{2}(S + S^T)\, x$

Since S is symmetric, $S = S^T$

$g^{(0)} = \nabla f(x^{(0)}) = \frac{1}{2} \cdot 2 S x = S x = \begin{bmatrix} 4 & 1 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 5 \\ 3 \end{bmatrix}$

$x^{(1)} = x^{[0]} - I^{-1} g^{(0)}$

$x(1) = \begin{bmatrix} 1 \\ 1 \end{bmatrix} - \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 5 \\ 3 \end{bmatrix}$

$= \begin{bmatrix} 1 \\ 1 \end{bmatrix} - \begin{bmatrix} 5 \\ 3 \end{bmatrix}$

$x(1) = \begin{bmatrix} -4 \\ -2 \end{bmatrix}$

$s_0 = x_1 - x_0$

$g_1 = \begin{bmatrix} 4 & 1 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} -4 \\ -2 \end{bmatrix}$

$= \begin{bmatrix} -16-2 \\ -4-4 \end{bmatrix} = \begin{bmatrix} -18 \\ -8 \end{bmatrix}$

$s_0 = \begin{bmatrix} -4 \\ -2 \end{bmatrix} - \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} -5 \\ -3 \end{bmatrix}$

$y_0 = g_1 - g_0 = \begin{bmatrix} -18 \\ -8 \end{bmatrix} - \begin{bmatrix} 5 \\ 3 \end{bmatrix} = \begin{bmatrix} -23 \\ -11 \end{bmatrix}$

$H_1 = \left[I - \frac{s_0 y_0^T}{y_0^T s_0} \right] H_0 \left[I - \frac{y_0 s_0^T}{y_0^T s_0} \right] + \frac{s_0 s_0^T}{y_0^T s_0}$

$s_0 y_0^T = \begin{bmatrix} 115 & 55 \\ 69 & 33 \end{bmatrix}$ -A

$y_0^T s_0 = 148$

$y_0 s_0^T = \begin{bmatrix} 115 & 69 \\ 55 & 33 \end{bmatrix}$ -B

$s_0 s_0^T = \begin{bmatrix} 25 & 15 \\ 15 & 9 \end{bmatrix}$ -D

$H_1 = \begin{bmatrix} 0.3567 & -0.291 \\ -0.291 & 0.8819 \end{bmatrix}$

(e.) $H_1 = H_0 + \dfrac{(s_0 - I y_0)(s_0 - I y_0)^T}{(s_0 - I y_0)^T y_0}$

$$s_0 - I y_0 = \begin{bmatrix} -5 \\ -3 \end{bmatrix} - \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} -23 \\ -11 \end{bmatrix}$$

$$= \begin{bmatrix} -5 \\ -3 \end{bmatrix} - \begin{bmatrix} -23 \\ -11 \end{bmatrix}$$

$$= \begin{bmatrix} 18 \\ 8 \end{bmatrix}$$

$(s_0 - I y_0)^T y_0 = -502$

$$H_1 = \begin{bmatrix} 0.3545 & -0.286 \\ -0.286 & 0.8725 \end{bmatrix}$$

Even though the same values are used for both the Hessian approximations are slightly different because BFGS applies rank 2 update while SR1 applies rank 1 update.

(f.) In the standard quasi-Newton secant equation information about 2 points is used to approximate the Hessian. But the curvature might change significantly through the search. That is why in the generalized equation, a generalized displacement vector & curvature vector are used.

Because more information you can give about the curvature more accurate the Hessian approximation is going to be.